

Clean Energy Nara Yoshino Power Plant (クリーンエナジー奈良 吉野発電所)

Summary of two Nara Prefectural Assembly committee site visits (22 Jan 2016 and 5 Aug 2015). Written in my own words; see the original PDFs for exact wording.

Basics

- Location: Masa, Ōyodo Town, Yoshino District, Nara Prefecture
- Operator: Clean Energy Nara Co., Ltd. (株式会社クリーンエナジー奈良)
- Type: woody biomass power plant, the first biomass plant in Nara Prefecture
- Completion ceremony: 17 December 2015 (commercial operation was planned from 15 December 2015, after test runs from November 2015)
- Output: 6,500 kW; about 5,800 kW can be sold after the plant's own consumption
- Site: about 8,000 m²
- Capital: ¥270 million; total project cost about ¥3.8 billion
- Annual electricity sales: about 43,000-44,000 MWh, roughly 12,000 households (enough for the households of Yoshino and Ōyodo)

Process

- Direct combustion: wood chips are burned in a boiler and the steam drives a turbine. The notes mention CO₂ leaving the boiler exhaust stack and a turbine driven by pressure. Gasification is not mentioned.
- The notes do not say which boiler type is used.
- The plant was not designed around heat use. Using waste heat for greenhouse cultivation was only an idea, and the operator said it would need technical study to confirm it wouldn't reduce electricity output.

Fuel

- Original plan: 50% unused forest wood (thinnings), 30% sawmill offcuts / general timber, 20% other wood.
- By Jan 2016, after about two years of collecting wood, unused wood made up about 80% of the stock.
- Unused wood is paid at the FIT (feed-in tariff) price for thinnings and general wood. Suppliers must provide the Forestry Agency's certification documents and a delivery note. Thinned wood must come with a municipal conformity notice, so only traceable wood gets the higher price.
- A council was formed to coordinate wood collection with forest cooperatives and timber associations.

Economics and local effects

- Electricity is sold at a fixed price for 20 years under the FIT scheme.
- Jobs: about 12 on-site workers and about 50 forestry workers supplying wood.

- Expected effects: more demand for low-grade (C and D grade) timber, support for timber prices, better forest water retention through thinning.
- Carbon: the operator argued CO2 from burning chips is offset by what the trees absorbed while growing.

Ash

- Expected to be 5-10 tonnes per day.
- Planned to test and analyze it after trial operation, then reuse it if safe (agriculture, compost, poultry farms for deodorizing, or processing into building materials).

Challenges and plans mentioned

- Securing fuel depends on forestry policy and on how many forestry workers remain.
- Forest roads need improving so timber can be hauled out without subsidies.
- Longer term, the operator floated setting up a local power retailer with Ōyodo and Yoshino towns, so locally generated power is used locally, once the initial sales contracts end.

Sources

- <https://www.pref.nara.jp/secure/154805/280122energy.pdf> (Jan 2016 visit)
- <https://www.pref.nara.jp/secure/147678/h270805nanbutobu.pdf> (Aug 2015 visit)
(Both currently return 403 / expired certificate for some users.)